

Recommendation Systems for Personalized Content Discovery

Netflix Prize Dataset · Collaborative Filtering & Matrix Factorization

100M+ Ratings

480K Users

17.7K Movies

99.96% Sparse

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Dataset Overview & The Sparsity Challenge

Netflix Prize Dataset · 100.5M ratings · 1999–2005

100.5M

Total Ratings

480,189

Unique Users

17,770

Movies

99.96%

Matrix Sparsity

User-Movie matrix (100x100 sample)
Fill rate: 0.04% -- 99.96% sparse



100×100 sampled submatrix — nearly all white (unseen)

Global mean rating

3.63 / 5 ★

58% ratings

4–5 stars (positivity bias)

Median ratings/user

5 (full dataset)

Model training

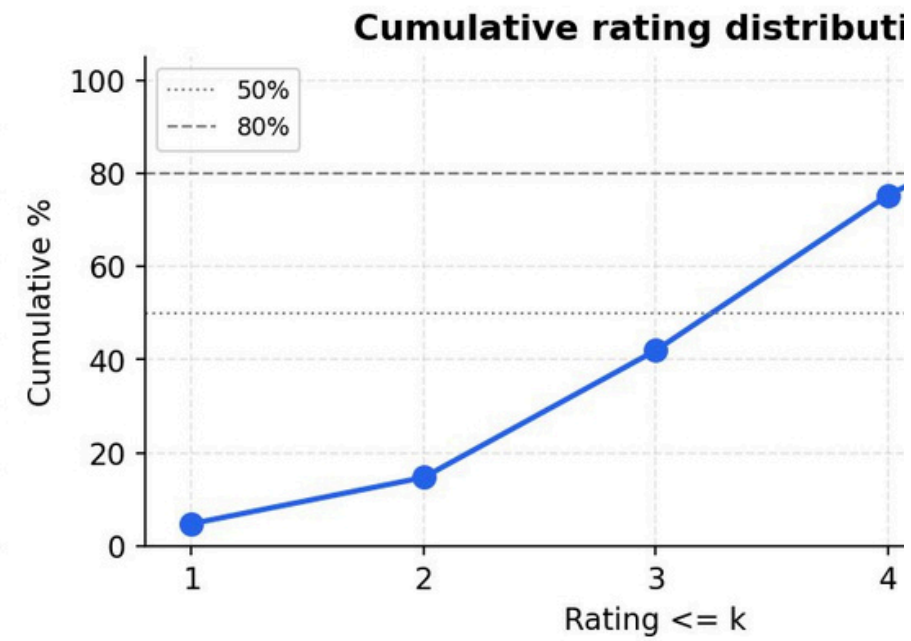
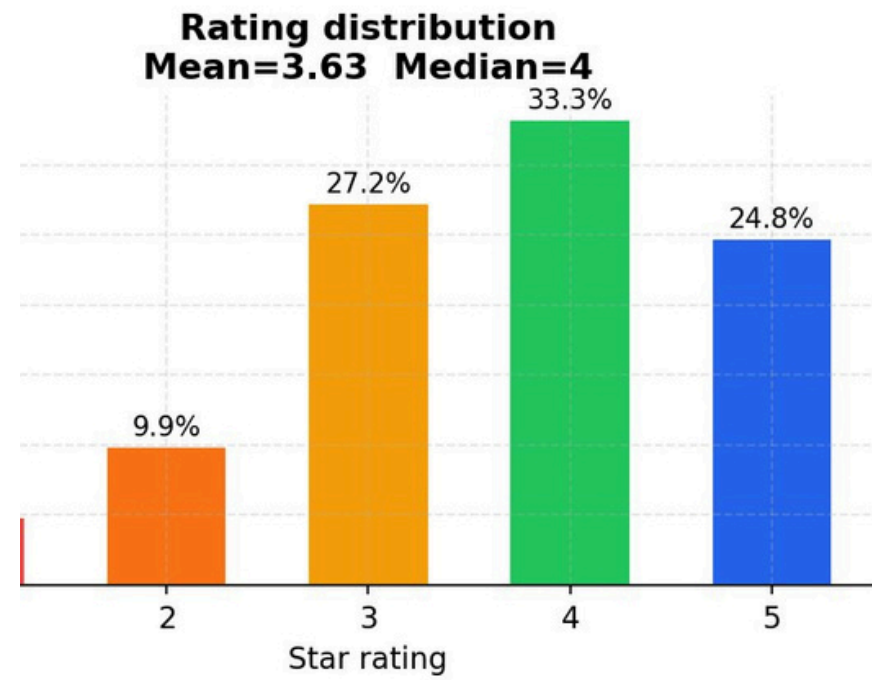
50,000 users · 11.7M ratings

Test set (probe.txt)

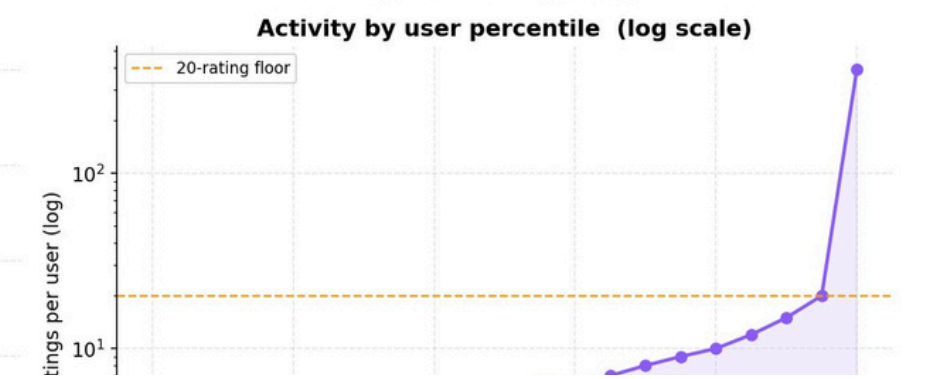
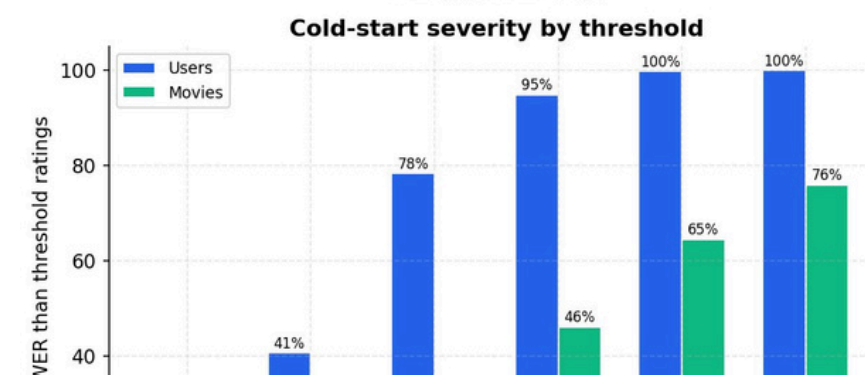
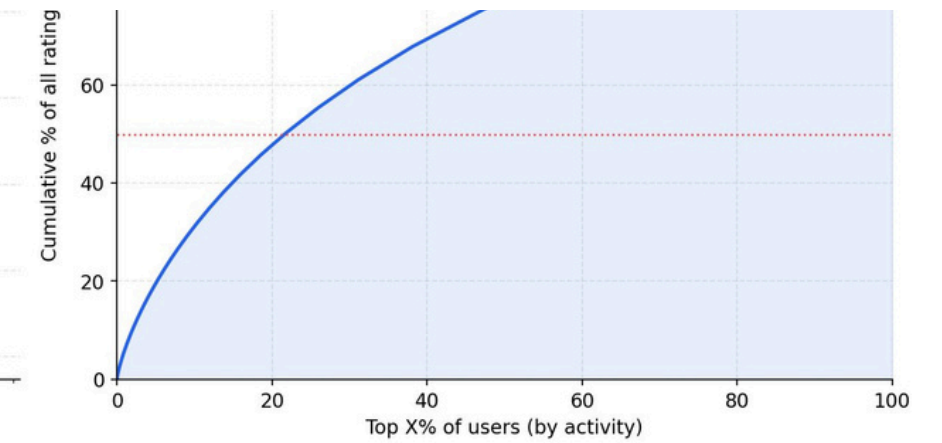
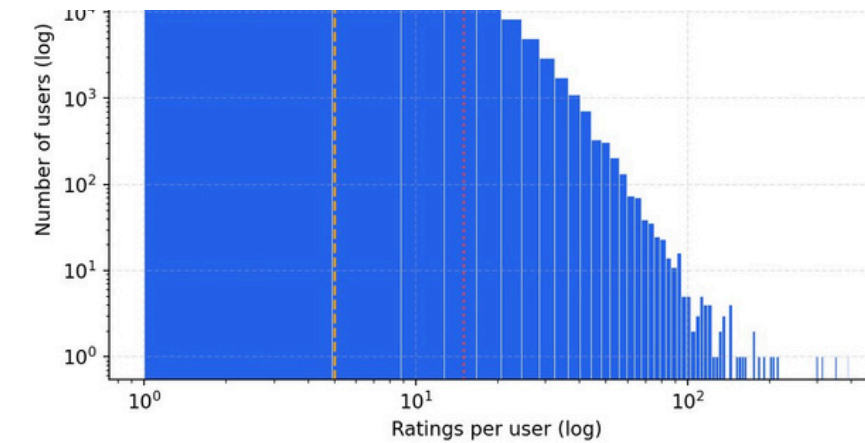
150,056 ratings · 48,682 users

Exploratory Data Analysis — Key Findings

Insight: 58% of ratings are 4-5 stars -- users self-select movies they expect to enjoy (positivity bias)



Rating Distribution: 58% are 4-5 ★ (positivity bias, mean=3.63)



User Activity: 95% of users have < 20 ratings (cold-start crisis)

99.96%

Sparse Matrix

Gini 0.834

Top 20% films = 88% ratings

$\sigma = 0.69$ ★

User rating bias spread

$r = 0.116$

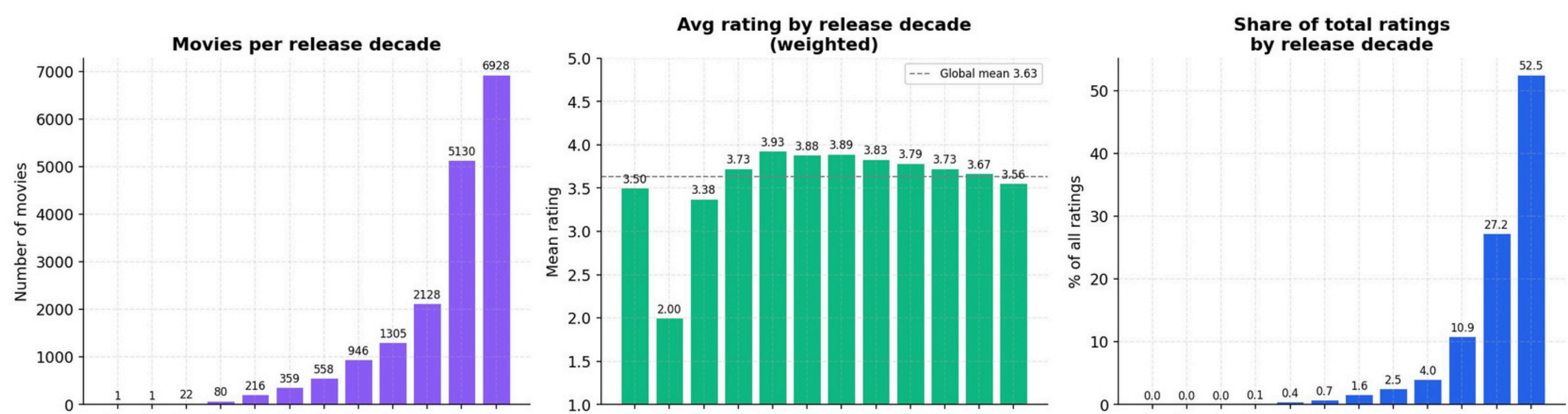
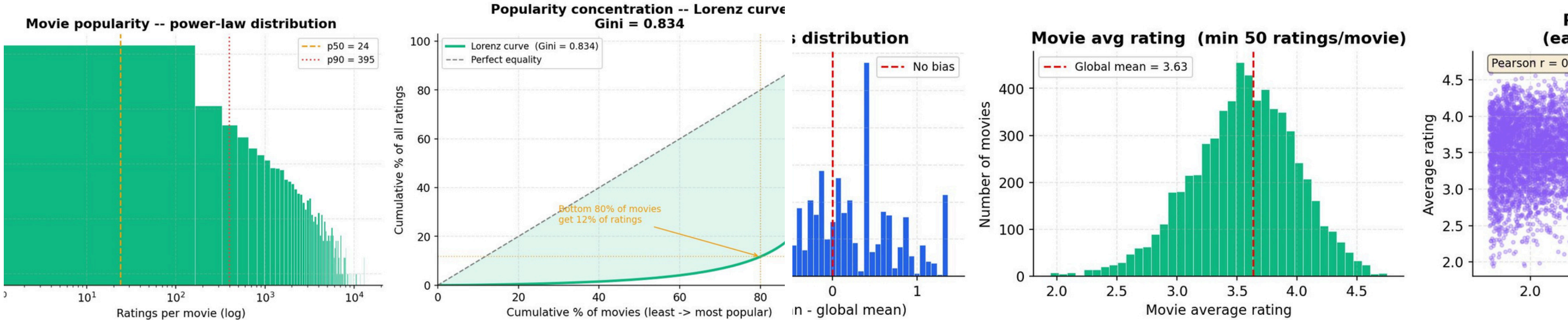
Popularity ≠ Quality

Key implication: SVD bias terms directly correct user + item rating tendencies — a critical advantage over plain CF. Relevance threshold ≥ 3.5 stars for MAP@10 is well-justified (lies above global mean).

EDA: Popularity Concentration & Rating Bias

Gini coefficient 0.834 · User bias $\sigma=0.69$ ★ · Survivor bias in classics
Insight: Top 20% of movies receive 88% of ratings. 28% of movies have < 10 ratings (long-tail).

Insight: Strong per-user and per-movie biases exist. SVD bias terms correct for this -- key advantage over plain CF.



Gini = 0.834

Top 20% of movies get 88% of ratings

$\sigma = 0.69$ ★

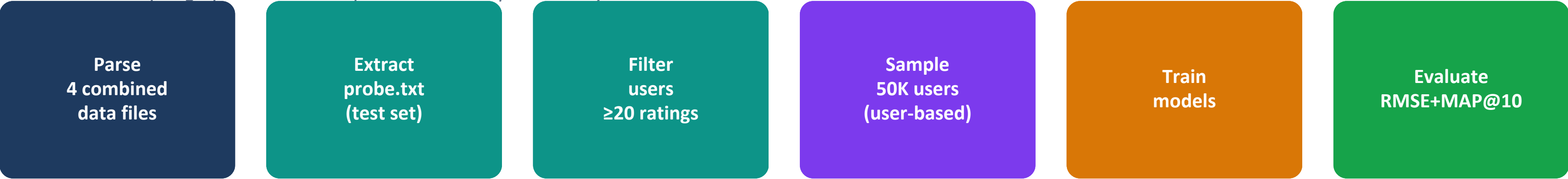
Strong per-user bias → SVD bias terms essential

Survivor bias

1930s films rated highest — only classics remain

Methodology: Pipeline & Model Design

User-based sampling · probe.txt test split · Three complementary models



Why user-based sampling? Row sampling destroys user histories (median 9 ratings/user → SVD worse than baseline, RMSE 1.048). User-based sampling preserves complete histories (median 120 ratings/user → SVD RMSE 0.9141, 6% gain).

Baseline

$$\mu + b_u + b_i$$

ALS-estimated user & item biases. No latent factors. Reference model.

SVD

$$\mu + b_u + b_i + p_u \cdot q_i$$

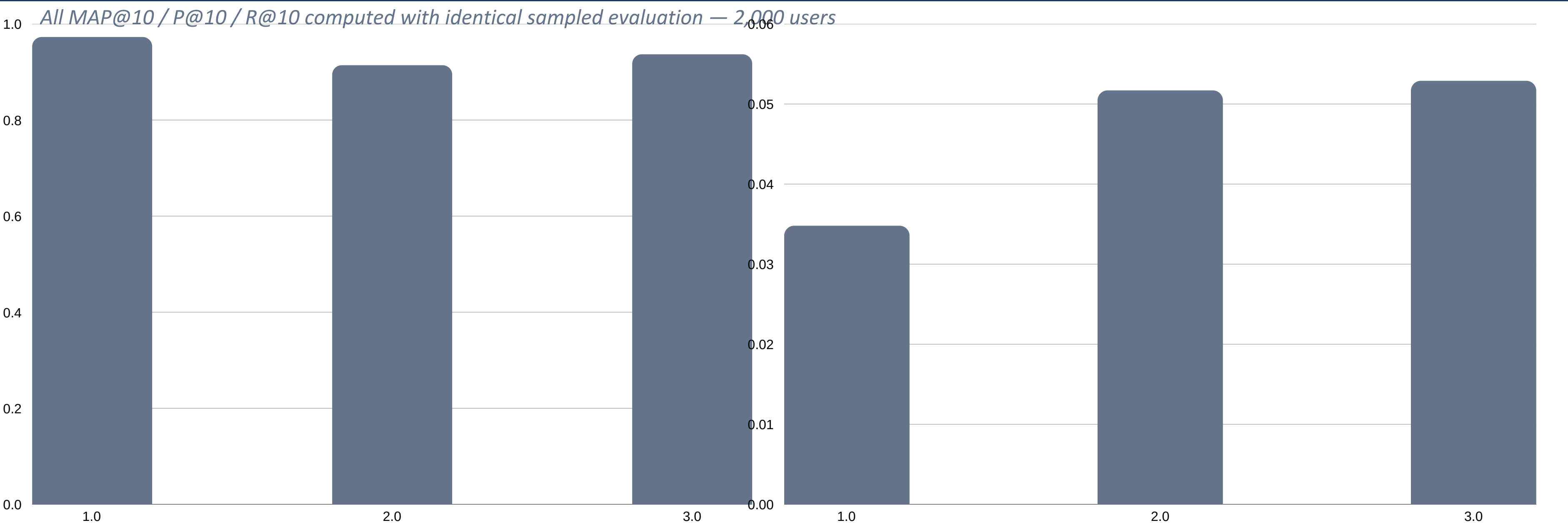
Matrix factorisation. 100 latent factors. 20 epochs SGD.
Captures taste compatibility via dot products.

Item-KNN

Cosine similarity
Top-2,000 movies

Neighbourhood-based CF. Provides "because you liked X" explanations. Restricted to top-2,000 for RAM.

Experimental Results



Model	RMSE	MAE	MAP@10	P@10	R@10
Baseline	0.9728	0.7640	0.0348	0.0186	0.0806
SVD ★	0.9141	0.7063	0.0517	0.0287	0.1292
Item-KNN †	0.9368	0.7215	0.0529	0.0307	0.1415

★ Best RMSE · † KNN restricted to top-2,000 movies; RMSE on popular-movie subset only

Recommendation Examples — Success & Cold-Start Failure

✓ SUCCESS | User 742175

293 ratings | avg 3.61★ | Loves: Matrix, Indiana Jones, Usual Suspects

Amelie (2001)	4.808
Lost: Season 1 (2004)	4.679
The West Wing: S1 (1999)	4.591
Arrested Development: S2	4.572
Whale Rider (2003)	4.558
U2: Live from Slane Castle	4.538
The Daily Show: Indecision 2004	4.535
The West Wing: S2 (1999)	4.583

World cinema + prestige TV + comedy — coherently reflects taste profile

✗ COLD-START FAIL | User 441484

36 ratings | Profile: children's films, animation, family

SVD (broken — predicts crime dramas)		Popularity fallback ✓
The Shield: S3	5.52	Pirates of Caribbean
Law & Order SVU	5.51	Forrest Gump
CSI: Season 3	5.24	The Green Mile
Passion of Christ	5.12	LotR: Two Towers

Problem: SVD predictions exceed 5.0 (out-of-scale) and suggest crime dramas to a children's animation fan.

Fix needed:
Content-based fallback using genre/year metadata, not just global popularity.

Key Insights & Future Improvements

KEY INSIGHTS

1. User-based sampling Preserves user histories → SVD gains 6% RMSE over baseline. Row sampling failed.

2. Sparsity is central 99.96% sparse matrix with Gini=0.834 popularity concentration defines all challenges.

3. Bias correction works $\sigma=0.69$ ★ per-user bias. SVD's explicit bias terms deliver consistent improvement.

4. Consistent evaluation Same candidate set essential. Different pool sizes make MAP@10 incomparable.

5. Explainability is free Item-KNN similarity matrix gives "because you liked" with no extra computation.

FUTURE IMPROVEMENTS

SVD++ Adds implicit feedback from which items a user rated. Typically +0.01–0.03 RMSE.

Temporal SVD Models taste drift over time using rating timestamps.

Neural CF (NCF) Deep networks capture non-linear user–item interactions.

Hybrid models Content-based features (year, genre) address cold-start for new items.

Full-scale KNN GPU-accelerated similarity (FAISS) removes 2,000-movie restriction.

Online learning Incremental parameter updates as new ratings arrive.